

# Double Integrals

ANSWER KEY



## Double integrals over a rectangle

- 1 Set up the double integral of  $f(x, y) = xy$  over the rectangle  $[0, 2] \times [0, 3]$  (integrate  $y$  first).  
 $\int_0^2 \int_0^3 xy \, dy \, dx.$
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## Iterated integrals

- 2 Evaluate  $\int_0^2 \int_0^3 xy \, dy \, dx.$   
Inner:  $\int_0^3 xy \, dy = x \cdot \frac{y^2}{2} \Big|_0^3 = \frac{9x}{2}.$  Outer:  $\int_0^2 \frac{9x}{2} \, dx = \frac{9}{4} x^2 \Big|_0^2 = 9.$
- 3 Evaluate  $\int_0^1 \int_0^2 (x^2 + y) \, dy \, dx.$   
Inner:  $\int_0^2 (x^2 + y) \, dy = x^2 y + \frac{y^2}{2} \Big|_0^2 = 2x^2 + 2.$  Outer:  $\int_0^1 (2x^2 + 2) \, dx = \frac{2}{3} x^3 + 2x \Big|_0^1 = \frac{8}{3}.$
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## Fubini's Theorem: order of integration

- 4 Verify Fubini's Theorem for  $f(x, y) = xy$  on  $[0, 1] \times [0, 2]$  by evaluating the integral in both orders.  
Order  $dy \, dx$ : inner  $= x \cdot \frac{y^2}{2} \Big|_0^2 = 2x,$  outer  $= \int_0^1 2x \, dx = 1.$  Order  $dx \, dy$ : inner  $= y \cdot \frac{x^2}{2} \Big|_0^1 = \frac{y}{2},$  outer  $= \int_0^2 \frac{y}{2} \, dy = 1.$  Both orders give 1.
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## Double integrals over general regions

- 5 Evaluate  $\int_0^1 \int_0^x (x + y) \, dy \, dx.$   
Inner:  $\int_0^x (x + y) \, dy = xy + \frac{y^2}{2} \Big|_0^x = x^2 + \frac{x^2}{2} = \frac{3x^2}{2}.$  Outer:  $\int_0^1 \frac{3x^2}{2} \, dx = \frac{1}{2} x^3 \Big|_0^1 = \frac{1}{2}.$
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## Polar coordinates

- 6 Convert  $\iint_D (x^2 + y^2) \, dA$  to polar coordinates, where  $D$  is the disk  $x^2 + y^2 \leq 4$ , and evaluate.  
 $x^2 + y^2 = r^2, dA = r \, dr \, d\theta.$  Inner:  $\frac{r^4}{4} \Big|_0^2 = 4.$  Outer:  $\int_0^{2\pi} 4 \, d\theta = 8\pi.$
- 7 Evaluate  $\iint_D xy \, dA$  where  $D$  is the quarter disk  $x^2 + y^2 \leq 1, x \geq 0, y \geq 0$ , using polar coordinates.  
 $xy = r^2 \sin \theta \cos \theta, dA = r \, dr \, d\theta, \theta \in [0, \frac{\pi}{2}], r \in [0, 1].$  Inner:  $\int_0^1 r^3 \, dr = \frac{1}{4}.$  Outer:  $\int_0^{\pi/2} \sin \theta \cos \theta \, d\theta = \frac{1}{2}.$
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## More iterated integral practice

- 8 Evaluate  $\int_0^1 \int_0^2 3x^2 y \, dy \, dx.$   
Inner:  $\int_0^2 3x^2 y \, dy = 3x^2 \cdot \frac{y^2}{2} \Big|_0^2 = 6x^2.$  Outer:  $\int_0^1 6x^2 \, dx = 2x^3 \Big|_0^1 = 2.$

9 Evaluate  $\int_0^1 \int_0^1 e^{x+y} dy dx$ .

Inner:  $\int_0^1 e^{x+y} dy = e^{x+y} \Big|_0^1 = e^{x+1} - e^x$ . Outer:

$$\int_0^1 (e^{x+1} - e^x) dx = [e^{x+1} - e^x]_0^1 = (e^2 - e) - (e - 1) = e^2 - 2e + 1 = (e - 1)^2.$$

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## The average value of a function

10 Find the average value of  $f(x, y) = xy$  over the rectangle  $[0, 2] \times [0, 4]$ , using  $\text{avg} = \frac{1}{\text{Area}} \iint_D f dA$ .

$$\text{Area} = 8. \int_0^2 \int_0^4 xy dy dx: \text{inner} = x \cdot \frac{y^2}{2} \Big|_0^4 = 8x; \text{outer} = \int_0^2 8x dx = 4x^2 \Big|_0^2 = 16. \text{Average} = \frac{16}{8} = 2.$$

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## More general-region integrals

11 Evaluate  $\int_0^1 \int_0^{1-x} (x + y) dy dx$  (the triangle with vertices  $(0, 0)$ ,  $(1, 0)$ ,  $(0, 1)$ ).

$$\text{Inner: } \int_0^{1-x} (x + y) dy = x(1 - x) + \frac{(1 - x)^2}{2} = \frac{1}{2} - \frac{x^2}{2}. \text{Outer: } \int_0^1 \left( \frac{1}{2} - \frac{x^2}{2} \right) dx = \left[ \frac{x}{2} - \frac{x^3}{6} \right]_0^1 = \frac{1}{2} - \frac{1}{6} = \frac{1}{3}.$$

12 Set up (don't evaluate)  $\iint_D x^2 y dA$  where  $D$  is the region between  $y = x^2$  and  $y = 2x$ , integrating  $y$  first (the curves meet at  $x = 0, 2$ ).

$$\int_0^2 \int_{x^2}^{2x} x^2 y dy dx.$$

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## Finding area with a double integral

13 Use a double integral to find the area of the region bounded by  $y = x^2$  and  $y = 2x$ .

$$\text{Area} = \int_0^2 \int_{x^2}^{2x} 1 dy dx = \int_0^2 (2x - x^2) dx = \left[ x^2 - \frac{x^3}{3} \right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}.$$

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## More polar-coordinate practice

14 Evaluate  $\iint_D \sqrt{x^2 + y^2} dA$  where  $D$  is the disk  $x^2 + y^2 \leq 9$ , using polar coordinates.

$$\sqrt{x^2 + y^2} = r, dA = r dr d\theta, \text{ so the integrand becomes } r^2. \int_0^{2\pi} \int_0^3 r^2 dr d\theta. \text{ Inner: } \frac{r^3}{3} \Big|_0^3 = 9. \text{ Outer:}$$

$$\int_0^{2\pi} 9 d\theta = 18\pi.$$

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## One more iterated integral

**15** Evaluate  $\int_0^2 \int_1^3 (2x + y) dy dx$ .

Inner:  $\int_1^3 (2x + y) dy = 2xy + \frac{y^2}{2} \Big|_1^3 = (6x + 4.5) - (2x + 0.5) = 4x + 4$ . Outer:

$$\int_0^2 (4x + 4) dx = 2x^2 + 4x \Big|_0^2 = 8 + 8 = 16.$$

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### A closing region-order problem

**16** Sketch the region for  $\int_0^1 \int_{\sqrt{y}}^1 x dx dy$  and rewrite it with the order of integration reversed.

The region is  $0 \leq y \leq 1$ ,  $\sqrt{y} \leq x \leq 1$ , i.e. between  $x = \sqrt{y}$  (or  $y = x^2$ ) and  $x = 1$ , for  $y$  from 0 to 1 —

equivalently  $0 \leq x \leq 1$ ,  $0 \leq y \leq x^2$ . Reversed:  $\int_0^1 \int_0^{x^2} x dy dx$ .

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### A final polar-coordinate application

**17** Find the volume under  $z = x^2 + y^2$  and above the disk  $x^2 + y^2 \leq 1$ , using  $V = \iint_D (x^2 + y^2) dA$  in polar coordinates.

$$x^2 + y^2 = r^2, dA = r dr d\theta. V = \int_0^{2\pi} \int_0^1 r^3 dr d\theta. \text{ Inner: } \frac{r^4}{4} \Big|_0^1 = \frac{1}{4}. \text{ Outer: } \int_0^{2\pi} \frac{1}{4} d\theta = \frac{\pi}{2}.$$

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### Setting up a volume integral over a rectangle

**18** Set up (don't evaluate) the double integral for the volume under  $z = 4 - x^2 - y^2$  and above the rectangle  $[0, 1] \times [0, 1]$ .

$$V = \int_0^1 \int_0^1 (4 - x^2 - y^2) dy dx.$$

**19** Evaluate the volume integral set up above.

$$\text{Inner: } \int_0^1 (4 - x^2 - y^2) dy = 4y - x^2y - \frac{y^3}{3} \Big|_0^1 = 4 - x^2 - \frac{1}{3} = \frac{11}{3} - x^2. \text{ Outer:}$$

$$\int_0^1 \left( \frac{11}{3} - x^2 \right) dx = \frac{11}{3}x - \frac{x^3}{3} \Big|_0^1 = \frac{10}{3}.$$

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### One last mass-density application

**20** A flat plate occupying the rectangle  $[0, 2] \times [0, 1]$  has density  $\rho(x, y) = x + y$  (mass per unit area). Set up the double integral for its total mass.

$$M = \int_0^2 \int_0^1 (x + y) dy dx.$$

- 21** Evaluate the total mass from the previous question.

Inner:  $\int_0^1 (x+y) dy = xy + \frac{y^2}{2} \Big|_0^1 = x + \frac{1}{2}$ . Outer:  $\int_0^2 \left(x + \frac{1}{2}\right) dx = \frac{x^2}{2} + \frac{x}{2} \Big|_0^2 = 2 + 1 = 3$ .

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### One more general-region setup and evaluation

- 22** Set up  $\iint_D y \, dA$  where  $D$  is the region bounded by  $y = 0$ ,  $y = x$ , and  $x = 4$ , integrating  $y$  first.

$$\int_0^4 \int_0^x y \, dy \, dx.$$

- 23** Evaluate the integral set up above.

Inner:  $\int_0^x y \, dy = \frac{y^2}{2} \Big|_0^x = \frac{x^2}{2}$ . Outer:  $\int_0^4 \frac{x^2}{2} \, dx = \frac{x^3}{6} \Big|_0^4 = \frac{64}{6} = \frac{32}{3}$ .

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### A last polar-coordinate integral

- 24** Evaluate  $\iint_D 1 \, dA$  (the area) where  $D$  is the half-disk  $x^2 + y^2 \leq 4$ ,  $y \geq 0$ , using polar coordinates.

$\theta \in [0, \pi]$ ,  $r \in [0, 2]$ .  $\int_0^\pi \int_0^2 r \, dr \, d\theta$ . Inner:  $\frac{r^2}{2} \Big|_0^2 = 2$ . Outer:  $\int_0^\pi 2 \, d\theta = 2\pi$  (matches  $\frac{1}{2}\pi(2)^2 = 2\pi$ , the known half-disk area formula).